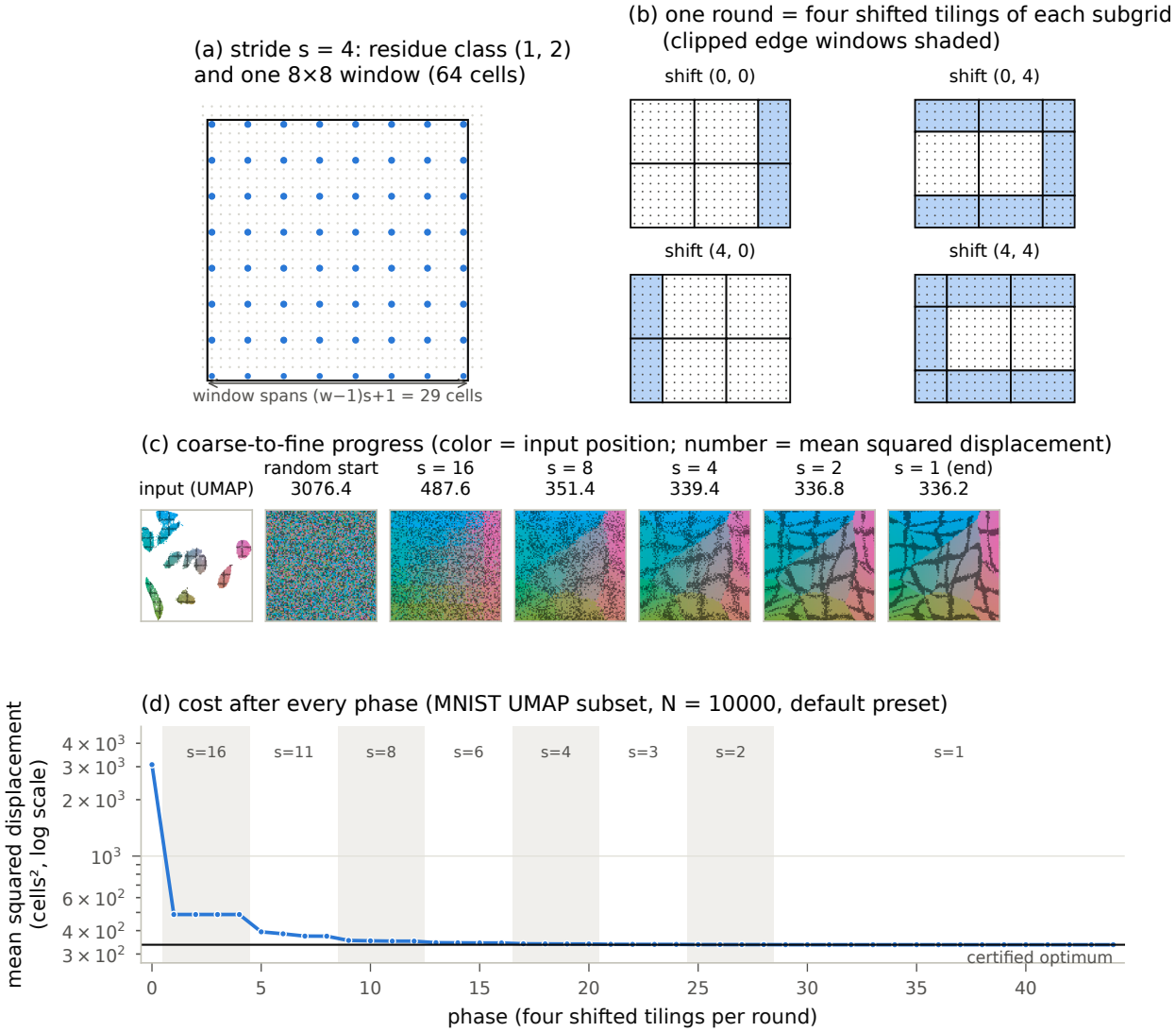

(a) stride s = 4: residue class (1, 2)
and one 8×8 window (64 cells)
window spans (w−1)s+1 = 29 cells
(b) one round = four shifted tilings of each subgrid
(clipped edge windows shaded)
shift (0, 0)
shift (0, 4)
shift (4, 0)
shift (4, 4)
(c) coarse-to-fine progress (color = input position; number = mean squared displacement)
input (UMAP)
random start
3076.4
s = 16
487.6
s = 8
351.4
s = 4
339.4
s = 2
336.8
s = 1 (end)
336.2
(d) cost after every phase (MNIST UMAP subset, N = 10000, default preset)
mean squared displacement
(cells², log scale)
4 × 10³
3 × 10³
2 × 10³
10³
6 × 10²
4 × 10²
3 × 10²
s=16
s=11
s=8
s=6
s=4
s=3
s=2
s=1
certified optimum
0
5
10
15
20
25
30
35
40
phase (four shifted tilings per round)